NIKHILA REDDY

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PROFESSIONAL SUMMARY

- AI/ML Engineer with 4+ years of experience developing machine learning, deep learning, NLP, Generative AI, and MLOps solutions across healthcare and energy analytics environments.
- Skilled in building predictive models, clinical document classification workflows, RAG applications, model monitoring frameworks, and scalable inference APIs using Python, Scikit-learn, TensorFlow, PyTorch, FastAPI, AWS, Spark, and SQL.
- Hands-on experience with LangChain, OpenAI API, Hugging Face, FAISS, ChromaDB, MLflow, text embeddings, prompt engineering, LLM evaluation, vector search, and retrieval optimization for modern AI applications.
- Strong background in data preprocessing, feature engineering, hyperparameter tuning, time-series modeling, anomaly detection, drift detection, model versioning, and production AI validation.
- Experienced in collaborating with data scientists, engineers, product owners, and business stakeholders to deliver AI solutions that improve review accuracy, reduce manual effort, and support faster decision-making.

TECHNICAL SKILLS

Programming & AI Development: Python, SQL, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Keras, OpenCV, FastAPI, Flask

Machine Learning & Predictive Modeling: Regression, Classification, Clustering, Random Forest, XGBoost, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Time-Series Modeling

Deep Learning & NLP: CNN, RNN, LSTM, Transformers, BERT, GPT Models, Hugging Face Transformers, SpaCy, NLTK, Text Classification, Named Entity Recognition

Generative AI & RAG: OpenAI API, LangChain, LlamaIndex, Prompt Engineering, Retrieval-Augmented Generation, LLM Evaluation, Prompt Testing, AI Response Validation

Vector Search & Embeddings: FAISS, ChromaDB, Text Embeddings, Vector Databases, Semantic Search, Similarity Search, Document Chunking, Retrieval Optimization

MLOps & Model Deployment: AWS SageMaker, MLflow, Experiment Tracking, Model Registry, Model Monitoring, Drift Detection, Model Versioning, TensorFlow Serving, Docker, Kubernetes, CI/CD for ML

Cloud & Data Platforms: AWS S3, AWS Lambda, AWS Glue, AWS Step Functions, Apache Spark, Apache Kafka

Databases, APIs & Reporting: PostgreSQL, REST APIs, Microservices, Power BI, Model Performance Dashboards

Collaboration & AI Governance: Git, JIRA, Agile, Scrum, PyTest, Unit Testing, Explainable AI, Bias Monitoring, Responsible AI

PROFESSIONAL EXPERIENCE

AI/ML Engineer — May 2024 - Present

HCA Healthcare | Indianapolis, IN

- Built clinical document classification models with Python, TensorFlow, PyTorch, and Scikit-learn to sort provider notes, intake forms, and operational records, improving AI-assisted review accuracy by 18%.
- Reduced manual documentation review effort by 32% by developing NLP pipelines with SpaCy, NLTK, BERT, GPT models, and Hugging Face Transformers to extract clinical terms, diagnosis patterns, and document intent.
- Created RAG workflows with LangChain, LlamaIndex, OpenAI API, FAISS, AWS S3, and FastAPI to help clinical and business teams search internal healthcare knowledge, cutting average information lookup time by 40%.
- Improved validated answer relevance by 27% through document chunking, text embeddings, metadata filters, prompt templates, and retrieval scoring for healthcare policy and knowledge content.
- Prepared model-ready datasets from 1M+ structured and unstructured healthcare records using Pandas, NumPy, SQL, Spark, and Kafka event inputs, improving data consistency for model training and batch scoring.
- Deployed real-time inference services using AWS SageMaker, FastAPI, REST APIs, Docker, Kubernetes, TensorFlow Serving, and CI/CD for ML workflows, enabling healthcare applications to receive sub-second prediction outputs.
- Implemented MLflow experiment tracking and model registry workflows to log parameters, metrics, artifacts, and deployment stages, improving traceability across model versions and release candidates.
- Added model monitoring checks with MLflow outputs, PyTest validation scripts, and performance logs to track drift, data quality, latency, and confidence-score changes before user impact.

Machine Learning Engineer — Jun 2020 - Oct 2022

Aguar Talent Care | Client: Abu Dhabi National Oil Company | India

- Developed predictive maintenance models with Python, Scikit-learn, Pandas, and NumPy using asset service history and equipment trend data, helping operations teams identify likely failures earlier.
- Increased asset risk detection by 24% through Random Forest, XGBoost, classification, and regression models that scored equipment health, ranked maintenance priority, and flagged operational exceptions.

- Cleaned and prepared 750K+ structured operational records with SQL, Python, and Pandas, improving training-data quality for asset risk modeling and batch prediction workflows.
- Engineered time-series features from rolling averages, lag variables, sensor trends, and service intervals, improving maintenance risk prediction accuracy by 19%.
- Reduced validation rework by 22% by tuning models with GridSearchCV, cross-validation, precision, recall, and F1-score analysis during model review cycles.
- Created anomaly detection workflows with clustering, statistical thresholds, and Python validation logic to identify unusual equipment readings for faster engineering investigation.
- Exposed asset risk scores through Flask-based REST APIs, reducing manual Python script dependency by 30% for operations and analytics users.
- Automated recurring data preparation and batch model scoring with Python scripts, SQL queries, and scheduled workflows, cutting manual reporting effort by 28% for analytics teams.

PROJECTS

Enterprise RAG Knowledge Assistant

- Built a document search assistant with Python, LangChain, LlamaIndex, OpenAI API, FAISS, ChromaDB, FastAPI, and AWS S3 to let users ask natural language questions across PDFs, FAQs, and policy files.
- Improved retrieval quality by applying text embeddings, document chunking, metadata filters, and semantic search logic, helping users reach relevant content faster without manual keyword searching.
- Added answer validation checks for groundedness, response relevance, and latency, reducing unsupported responses and making the assistant more dependable for internal knowledge lookup.

Model Monitoring and Drift Detection Framework

- Developed a reusable model monitoring framework with Python, MLflow, Pandas, Scikit-learn, and SQL outputs to track prediction quality, missing values, drift signals, and confidence-score changes.
- Created Power BI model health dashboards showing accuracy trends, drift movement, failed validation checks, and retraining indicators, giving teams clearer visibility into model behavior.
- Defined threshold rules, model-version tracking, PyTest validation checks, and rollback notes, helping teams manage ML models with stronger governance and less manual follow-up.

Medical Image Classification and Triage Model

- Built an image classification pipeline with TensorFlow, Keras, CNNs, OpenCV, and image augmentation to classify medical image patterns and support faster prioritization of review queues.
- Improved validation performance by using transfer learning, hyperparameter tuning, confusion matrix review, and class imbalance handling, helping reduce false-negative risk during model testing.
- Packaged the trained model through FastAPI, Docker, and CI/CD for ML steps so applications could submit image inputs and receive prediction scores through a reusable inference service.

EDUCATION

Master of Science in Computer Science **Jan 2023 - May 2024**
University of Central Missouri | Warrensburg, MO

Bachelor of Science in Computer Science **Jun 2018 - May 2022**
Sri Dattha Institute of Engineering and Science | Hyderabad, India

CERTIFICATIONS

- Machine Learning Specialization - **Coursera**
- Generative AI with Large Language Models - **Coursera**
- Deep Learning Specialization - **Coursera**
- Natural Language Processing Specialization - **Coursera**
- MLOps Tools: MLflow and Hugging Face - **LinkedIn Learning**
- Building Generative AI Applications with LangChain - **Coursera**